

SAR-PACT: Tokenization Energy-Aware Complexity Allocation for Practical Learned SAR Image Compression

Zach Button, Paras Maharjan, Abdul Malik Badshah, Zhu Li
School of Science and Engineering
University of Missouri-Kansas City
Kansas City, MO 64110
(zb7df, paras.maharjan, ambnrt, lizhu)@umsystem.edu

Abstract—Compression of Synthetic Aperture Radar (SAR) imagery presents several challenges compared to passive sensor images, including dissimilar statistics of pixel values in the complex signal domain, high occurrence of speckle noise, and the need to preserve both Amplitude and Phase components with high fidelity. Existing Learned Image Compression (LIC) methods deliver strong rate-distortion performance on SAR images vis-a-vis traditional codecs, but they are often impractical for resource-constrained systems or low latency requirements. In contrast, traditional codecs trade compression performance for efficiency gains. We present Partial convolution-based Adaptive Compression in the Token domain (PACT), a lightweight neural compression architecture that incorporates Discrete Cosine Transform (DCT)-based tokenization to generate token feature channels with a known energy distribution and subsequent energy-driven partial convolution blocks in the learned compression pipeline. This significantly reduces the complexity of the latent representation learning, and when optimized for complex-valued SAR imagery, PACT achieves Bjøntegaard Delta (BD)-Rate improvements of 20% and 28% over VVC in Amplitude and Phase recovery, respectively. This is achieved with only 20.34 kMACs/pixel, a more than 20x reduction compared to Efficient Learned Image Compression (ELIC), demonstrating that SAR-optimized learned compression can achieve state-of-the-art quality without sacrificing real-world practicality.

Index Terms—SAR image, image compression, transform-domain processing

I. INTRODUCTION

Synthetic Aperture Radar (SAR) is an active, coherent imaging modality that forms images by transmitting microwave signals and measuring their backscatter from the Earth’s surface. Unlike optical sensors, which depend on visible illumination, SAR operates independently of lighting conditions and can penetrate atmospheric obscurants such as clouds, fog, smoke, and vegetation, enabling robust imaging in challenging environments [1], [2]. As a coherent system, SAR captures both the intensity and Phase of the returned signal. The resulting measurements are represented as complex-valued data in the In-phase (I) and Quadrature (Q) components, which can be converted into Amplitude (A) and Phase (ϕ) representations [3]:

$$A = \sqrt{I^2 + Q^2} \quad (1)$$

$$\phi = \arctan(Q/I) \quad (2)$$

The Amplitude reflects the strength of the backscattered signal, often exhibiting speckle due to coherent processing, while the Phase encodes fine-grained structural information related to surface geometry and elevation. Although single-look Phase is noisy and difficult to interpret, multi-pass Phase measurements enable critical use cases like change detection, so Phase reconstruction is an equally important task to Amplitude recovery for some applications.

SAR systems are commonly deployed on resource-constrained platforms including UAVs, drones, and satellites, where onboard computation, memory, and communication bandwidth are limited [4]. Crucially, these platform constraints not only motivate the need for compression, but also impose strict requirements on computational efficiency. In practical systems, encoding must be lightweight and memory-efficient to operate onboard in real time, while minimizing decoder latency is critical for rapid response scenarios such as disaster management.

We propose Partial convolution-based Adaptive Compression in the Token domain (PACT), a lightweight architecture that concentrates model capacity where signal energy is highest. In this work, we further optimize this architecture for SAR imagery using SAR-specific objective functions and a two-stage training procedure. The result, SAR-PACT, is a system that achieves competitive SAR reconstruction quality relative to state-of-the-art (SOTA) learned compression methods at a fraction of their computational cost.

The main contributions of this work are as follows:

- Block DCT tokenization and Energy-Guided Grouped Convolution: A novel first-layer encoding design that allocates convolutional filters proportionally to input channel energy, enabling an 8x complexity reduction with little impact on compression performance.

- SAR-Specific Loss Function: A composite rate-distortion objective that jointly optimizes reconstruction fidelity in both the complex I/Q domain and the Amplitude domain.
- Two-Stage Training Scheme: A staged training procedure which optimizes exclusively for I/Q fidelity before fine-tuning for Amplitude, improving Amplitude reconstruction without sacrificing Phase quality.

II. LITERATURE REVIEW

A. Traditional Compression

Optical image compression has been extensively studied, yielding a range of established standards including JPEG [5], JPEG2000 [6], SPIHT [7], BPG [8], and intra-mode variants of HEVC [9], [10] and VVC [11], [12]. CPU-friendly implementations have since been developed, including VVenC [13], and VVdeC [14], demonstrating effectiveness on resource-constrained devices. Several works have adapted these traditional optical methods for SAR image compression [15]–[22], but they introduce blocking, ringing, and blurring artifacts that become particularly pronounced in SAR Phase data. More broadly, these methods are fundamentally ill-suited to SAR data, due to its high dynamic range, extreme noise levels, and the sensitivity of the Phase component to compression-induced distortion.

B. Learning-based Compression

Learned image compression methods based on the variational autoencoder (VAE) framework introduced by Ballé et al. [23], [24] and extended by Minnen et al. [25], [26] have achieved strong rate-distortion performance on natural imagery, motivating their adaptation to SAR compression [27]–[30]. However, general-purpose learned codecs perform poorly on SAR data, with particular challenges in handling low information content, minimal spatial redundancy, and high noise levels for which the standard architectures were not designed. SAR-specific end-to-end compression methods have been proposed to address these limitations, leveraging VAE-based architectures to improve rate-distortion performance on both Amplitude and Phase reconstruction, as well as other downstream tasks [31], [32]. Despite these advances, the computational complexity of such approaches remains prohibitive for real-time deployment on resource-constrained platforms such as drones and UAVs.

Computational efficiency has emerged as a critical third axis of optimization in learned image compression, complementing the traditional rate-distortion trade-off [33]–[36]. Recent work has begun to explicitly address the computational burden of learned image compression by incorporating architectural and operational modifications aimed at reducing complexity [37]–[41]. Popular approaches include modifying the context model for parallelization or progressive coding [42]–[45], as well as utilizing an asymmetric structure to optimize decoder efficiency at the expense of encoder complexity [46], [47]. Of particular relevance to our work, Chen et al. [48] propose a novel partial convolution layer as part of the FasterNet architecture, reducing redundant computation by selectively

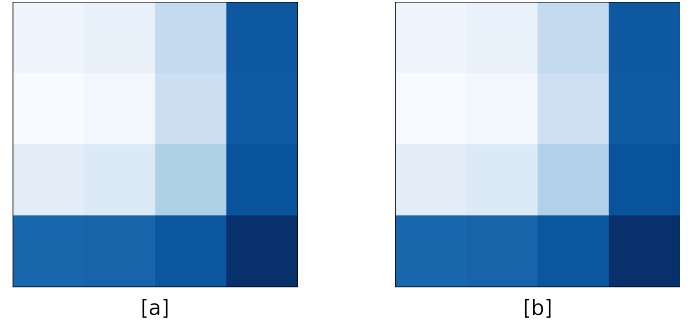


Fig. 1. Visualization of average energy per DCT-token channel, resulting from 4×4 2D DCT. [a] In-phase (I), [b] Quadrature (Q)

applying convolutional operations to a subset of channels. He et al. [49] introduce an Asymmetric Hyperprior Transform (AHT) with an Energy Allocation (EA) loss that routes higher-energy latent channels through deeper network branches, reducing redundant computation on low-energy channels. These approaches highlight a broader trend toward structure-aware and computation-efficient design, and they motivated further exploration of methods that allocate resources according to signal characteristics while maintaining competitive compression performance.

III. PROPOSED METHOD

This section describes the design and implementation of the proposed PACT framework and its adaptation to SAR imagery (SAR-PACT), including the preprocessing pipeline, model architecture, training configuration, and evaluation metrics.

A. Dataset

This study utilizes the National Geospatial-Intelligence Agency (NGA) SAR dataset [50]. While this dataset contains quad-polarization imagery, only the horizontally transmitted and received (HH) polarization channel is used throughout this work. The dataset consists of two large-format acquisitions of the same scene, each approximately 3800×6600 pixels. From the first image, 10,000 non-overlapping patches of size 256×256 pixels were extracted for training and validation. The second capture was held out as a single high-resolution test image. Each of the test image's dimensions was expanded to a multiple of 64 via reflection padding to enable PACT's required level of downsampling. This resulted in an input spatial size of 3904×6656 pixels.

To constrain the dynamic range and mitigate the influence of localized high-intensity spikes, a clipping operation was applied to both the I and Q channels. Pixel values were clipped to the interval $[-\sigma_{cut}, +\sigma_{cut}]$, where the threshold σ_{cut} was selected such that no more than 4% of pixels across both channels were affected. By limiting the dynamic range, we suppressed the impact of extreme intensity values while preserving the information content of the primary SAR signal.

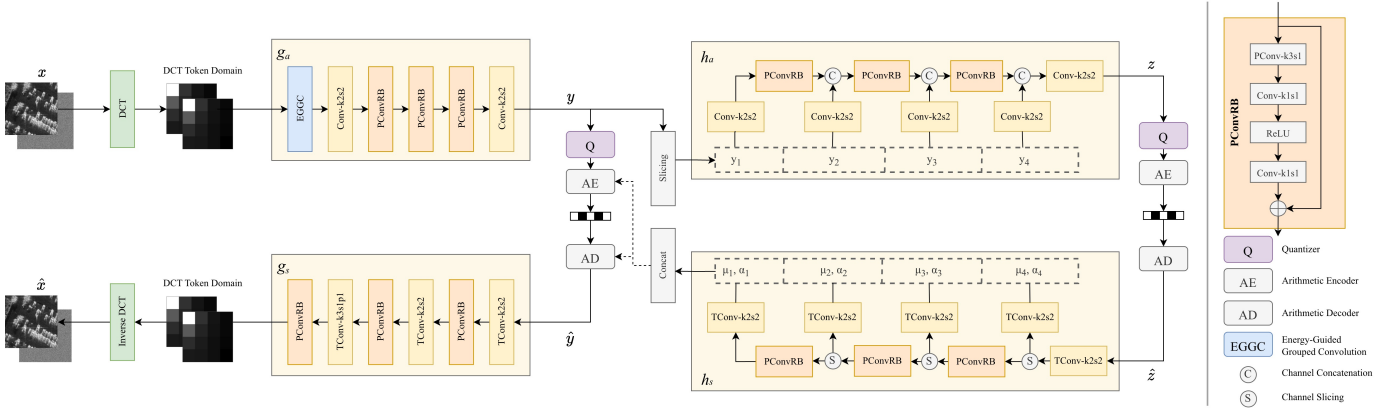


Fig. 2. Diagram illustrating our proposed end-to-end architecture, utilizing the standard VAE design with DCT preprocessing and energy-aware transform networks.

B. Token-Domain Processing

The I and Q components of complex-valued SAR imagery exhibit a band-limited nature arising from the inherent sampling constraints of SAR sensors, which can be restructured using a Discrete Cosine Transform (DCT). This DCT representation reveals that signal energy is predominantly concentrated in the low-frequency coefficients, with high-frequency components approaching negligible magnitudes. This behavior is a direct consequence of frequency-domain zero-padding employed by high-resolution SAR systems to achieve oversampling.

This spectral structure is exploited directly in the PACT preprocessing stage, inspired by Maharjan et al. [31]. Their approach applied DCT preprocessing to the Efficient Learned Image Compression (ELIC) framework [42], hereafter referred to as ELIC-DCT, achieving SOTA performance at lower complexity. A 4×4 2D blockwise DCT is applied independently to the I and Q channels, converting the spatial-domain signal into the transform domain. The resulting DCT coefficients are then rearranged via pixel unshuffling, reducing the spatial dimensions from (H, W) to $(\frac{H}{4}, \frac{W}{4})$ and expanding the channel dimension from 1 to 16. We refer to this rearranged DCT representation as the token domain, where each spatial position carries a vector of DCT frequency coefficients corresponding to a local 4×4 block. Fig. 1 illustrates the average energy of DCT-token channels $\{DC, AC_1, \dots, AC_{15}\}$ in raster order. Finally, these DCT-tokenized representations of the I and Q channels are concatenated along the channel dimension, creating the $(32, \frac{H}{4}, \frac{W}{4})$ input to the main analysis transform network.

This preprocessing offers two advantages. First, it rearranges the input image into a predictable energy structure, which can be exploited by allocating model capacity to higher-energy channels. Second, the token-domain representation increases the effective receptive field of the analysis transform by a factor of 16 pixels, enabling the model to capture longer-range spatial dependencies. Additionally, this transformation is lossless with negligible complexity, so these benefits are available at virtually no cost.

C. PACT Architecture

PACT extends the standard learned image compression framework with two targeted modifications designed to exploit a predictable energy structure and minimize computational overhead: an adaptive grouped convolution layer and an asymmetric hyperprior design. Together, these components form an energy-aware architecture that achieves strong rate-distortion performance at dramatically reduced complexity relative to existing learned SAR compression methods. The full PACT architecture is shown in Fig. 2.

1) *Energy-Guided Grouped Convolution*: The DCT preprocessing applied to the I and Q channels produces a 32-channel tokenized representation with a reduced spatial resolution of $\frac{H}{4} \times \frac{W}{4}$ pixels, which serves as input to the first layer of the analysis transform network. This first layer implements the Energy-Guided Grouped Convolution (EGGC). Grouped convolution layers with uniform filter allocation were pioneered in early deep learning architectures like AlexNet [51] and ResNeXt [52], for the purposes of parallel processing and increased model capacity. Grouped convolutions with a fixed non-uniform filter allocation have also been proposed to reduce complexity while maintaining model performance [53]. Our novel EGGC layer extends these ideas by dynamically allocating filters according to the calculated energy in each channel group.

The number of multiply-accumulate operations required by a grouped convolution layer is given by

$$MAC_G = \frac{K_H \times K_W \times H_{out} \times W_{out} \times C_{in} \times C_{out}}{G} \quad (3)$$

where $K_H \times K_W$ is the filter kernel size, $H_{out} \times W_{out}$ are the output spatial dimensions, C_{in} and C_{out} are the input and output channels, and G is the number of evenly-sized groups. This calculation holds whether the C_{out} filters are evenly or unevenly allocated across groups.

Channel energy is defined as variance and pre-computed across the entire training set at initialization, capturing and summarizing the dataset's spectral statistics. The 32 input channels are sorted by descending energy and partitioned

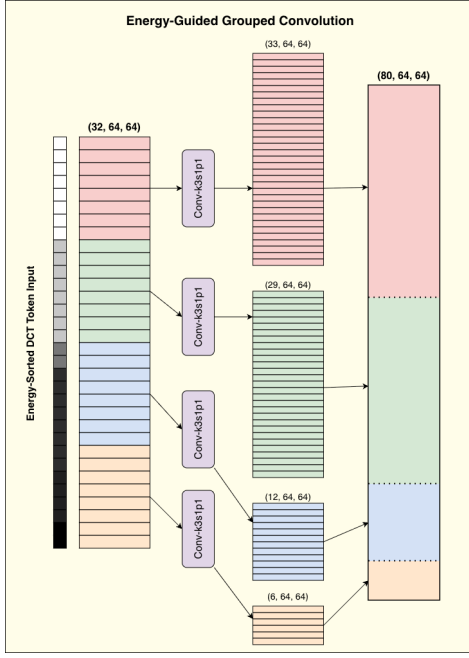


Fig. 3. Diagram illustrating the Energy-Guided Grouped Convolution layer architecture. Conv-k3s1p1 refers to a 3×3 convolution with stride 1 and padding 1.

into 8 equal groups of 4 channels each. To evaluate the effect of different channel grouping schemes, we performed an ablation study in Section IV-C, demonstrating that our chosen configuration provides a valuable balance between compression performance and complexity.

The proportion of total energy contained in each group determines its allocation of output filters from a total budget of 80, subject to a minimum allocation of 1 filter per group. The 8 groups are processed by independent 3×3 convolutional layers with stride 1 and padding 1, preserving spatial dimensions, and the 8 outputs are concatenated along the channel dimension to produce an output tensor of shape $(80, \frac{H}{4}, \frac{W}{4})$. This tensor is passed directly into a standard convolutional layer with learned weights, without the need for channel mixing. A simplified version of the EGGC architecture is shown in Fig. 3 for an input image with 256×256 pixels and $G = 4$.

2) *Asymmetric Hyperprior Transform*: The hyperprior network follows the AHT design proposed by He et al. [49], adapted to a higher-dimensional latent tensor for PACT. The 320 latent channels are partitioned into 4 sequential groups of 80 channels each. An Energy Allocation (EA) loss penalizes each group proportionally to its energy, encouraging the groups to converge to a naturally energy-sorted arrangement in which the first 80 channels carry the most energy and the final 80 channels carry the least. All four groups are passed through independent downscaling convolutional layers, then progressively aggregated with increasing depth. This progressive aggregation ensures that higher-energy groups receive more convolutional processing, concentrating model capacity where signal information is most dense. The hyperprior decoder mirrors this asymmetry, outputting per-element means and

scales that parameterize independent Gaussian distributions consumed by the arithmetic coder.

D. Loss Function

SAR-PACT is trained by minimizing a composite rate-distortion loss that jointly optimizes bitrate, reconstruction fidelity in both the I/Q and Amplitude domains, and the energy allocation of latent channel groups:

$$Loss = R + \lambda(\alpha D_{IQ} + (1 - \alpha) D_{Amp}) + \gamma \mathcal{L}_{EA} \quad (4)$$

where R is the estimated bitrate in bits per pixel, D_{IQ} and D_{Amp} are domain-specific distortion terms, $\mathcal{L}_{EA}$ is the EA loss, λ is the standard Lagrangian rate-distortion multiplier, α controls the relative weight of the two distortion terms, and γ is a scaling parameter for the EA loss term.

D_{IQ} is defined as an equally-weighted combination of L1 loss ($\mathcal{L}_{L1}$) and Structural Similarity (SSIM) loss ($\mathcal{L}_{SSIM}$), where $\mathcal{L}_{L1} = \mathbb{E}[|x - \hat{x}|]$ and $\mathcal{L}_{SSIM} = 1 - SSIM(x, \hat{x})$. This approach encourages both pointwise accuracy and local structural coherence in the reconstructed I/Q signal.

While Peak Signal-to-Noise Ratio (PSNR) is often used to measure reconstruction quality, it does not properly account for the multiplicative nature of the speckle noise prevalent in SAR imagery. Instead, we measure Amplitude fidelity via Signal-to-Quantization-Noise Ratio (SQNR). Let x_i and $\hat{x}_i$ denote the original and reconstructed Amplitude values at pixel i . The local signal power and noise power at pixel i are defined as follows:

$$S_i = \mathbb{E}_i[x^2] = \frac{1}{k^2} \sum_{j \in R_i} x_j^2 \quad (5)$$

$$N_i = \mathbb{E}_i[(x - \hat{x})^2] = \frac{1}{k^2} \sum_{j \in R_i} (x_j - \hat{x}_j)^2 \quad (6)$$

where R_i denotes the $k \times k$ neighborhood centered at pixel i . The image-level SQNR is the mean of their ratio across all M pixels and expressed in decibels (dB):

$$SQNR = 10 \cdot \frac{1}{M} \sum_{i=1}^M \log_{10} \left(\frac{S_i}{N_i} \right) \quad (7)$$

Since S_i is independent of model parameters, it is constant with respect to optimization and can be dropped from the objective function. The remainder is negated to form a simplified minimization objective, yielding an SQNR-derived Amplitude distortion loss:

$$D_{Amp} = \frac{1}{M} \sum_{i=1}^M \log_{10} (N_i) \quad (8)$$

where $N_i = \mathbb{E}_i[(x - \hat{x})^2]$ is the local Mean Squared Error (MSE) over the $k \times k$ neighborhood centered at pixel i . We utilized local neighborhoods with size $k = 5$ for all experiments.

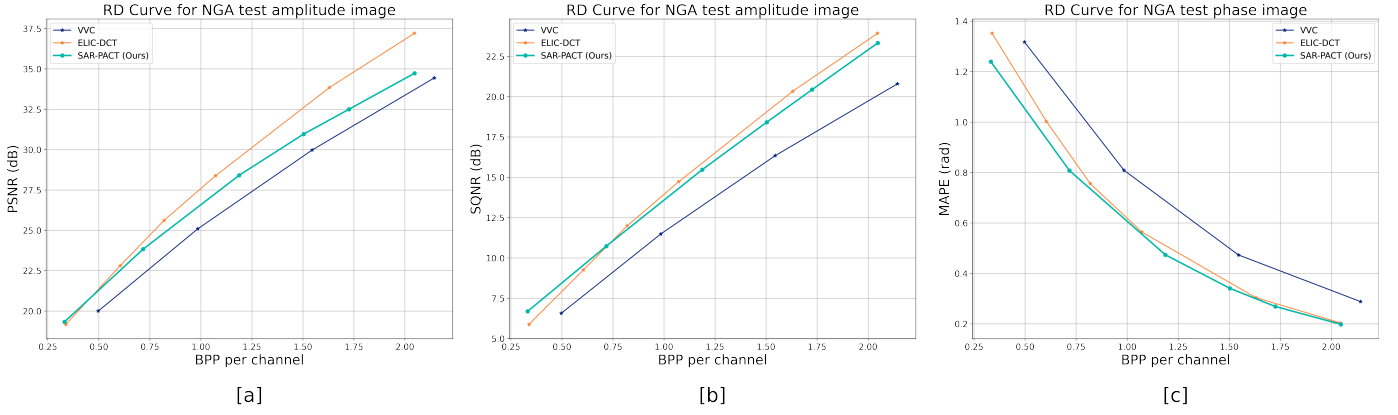


Fig. 4. RD curve visualization for NGA dataset, showing [a] BPP vs PSNR, [b] BPP vs SQNR, and [c] BPP vs MAPE.

The EA loss encourages the four latent channel groups to self-organize with decreasing energy from group 1 to group 4, enabling energy-driven channel routing in the hyperprior network. It is formulated as a weighted average of Mean Absolute Deviation across the four groups:

$$\mathcal{L}_{EA} = \sum_{i=1}^g \left(\beta_i \cdot \frac{1}{n_i} \sum_{j=1}^{n_i} |y_j - \mu_i| \right) \quad (9)$$

E. Training Details

λ is varied across training runs to obtain multiple rate-distortion operating points, following standard practice in learned image compression research. γ is fixed at 1.0 throughout training. Model training is executed in two stages: Stage 1 trains for 250 epochs with $\alpha = 1$, optimizing exclusively for I/Q fidelity, and Stage 2 trains for an additional 100 epochs with $\alpha = 1.7 \times 10^{-3}$, to fine-tune for Amplitude reconstruction.

SAR-PACT model training was performed using a batch size of 32, an Adam optimizer [54], and an initial learning rate of 1×10^{-3} with cosine annealing. SAR-PACT and ELIC-DCT were each evaluated across six operating points. VVC was tested via VTM 23.14 [55] with 12-bit quantization across four quantization parameter (QP) values. All methods were benchmarked on a Rocky Linux system with 8 CPU cores and one NVIDIA L40S GPU.

F. Performance Metrics

Our primary metric for measuring Amplitude distortion is SQNR (7), which reflects the local signal-to-noise characteristics of SAR Amplitude data. PSNR is also reported to facilitate direct benchmarking against learned image compression literature that does not report SQNR:

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right) \quad (10)$$

where MAX_I denotes the maximum possible value of an image and $MSE = \mathbb{E}[(x - \hat{x})^2]$.

Phase fidelity is evaluated using Mean Absolute Phase Error (MAPE) over all M pixels:

$$MAPE = \frac{1}{M} \sum_{i=1}^M |x_i - \hat{x}_i| \quad (11)$$

where x_i and $\hat{x}_i$ are the original and reconstructed Phase values for pixel i .

Bitrate is reported as bits per pixel (BPP) per channel, where $F_{\hat{y}}$ represents the compressed file size in bytes and each image contains $C = 2$ channels and $H \times W$ pixels:

$$BPP \text{ per channel} = \frac{8 \times F_{\hat{y}}}{C \times H \times W} \quad (12)$$

To evaluate compression performance across a range of bitrates, a rate-distortion (RD) curve is generated. This curve represents the trade-off between compression rate and reconstruction quality. For Amplitude, RD curves are shown for both PSNR and SQNR, where a larger area under the curve (AUC) represents better performance. For the Phase component, an RD curve is shown for MAPE, where a smaller AUC represents better performance. To summarize a method's performance across the entire RD curve, we employ the Bjøntegaard Delta Rate (BD-Rate) [56], which estimates the relative difference in bitrate between two codecs at equivalent levels of distortion. We calculate all BD-Rates relative to a VVC baseline.

Additionally, model complexity is measured in thousands of multiply-accumulate operations per pixel (kMACs/px).

IV. EXPERIMENTAL RESULTS

A. Rate-Distortion Results

We evaluated the RD performance of each method on the second scene in the NGA dataset. Fig. 4 shows the RD curves for PSNR, SQNR, and MAPE. SAR-PACT achieves 20.46% and 28.45% improvements in BD-rate compared to VVC in terms of Amplitude SQNR and MAPE, respectively. However, this RD performance is not accompanied by the level

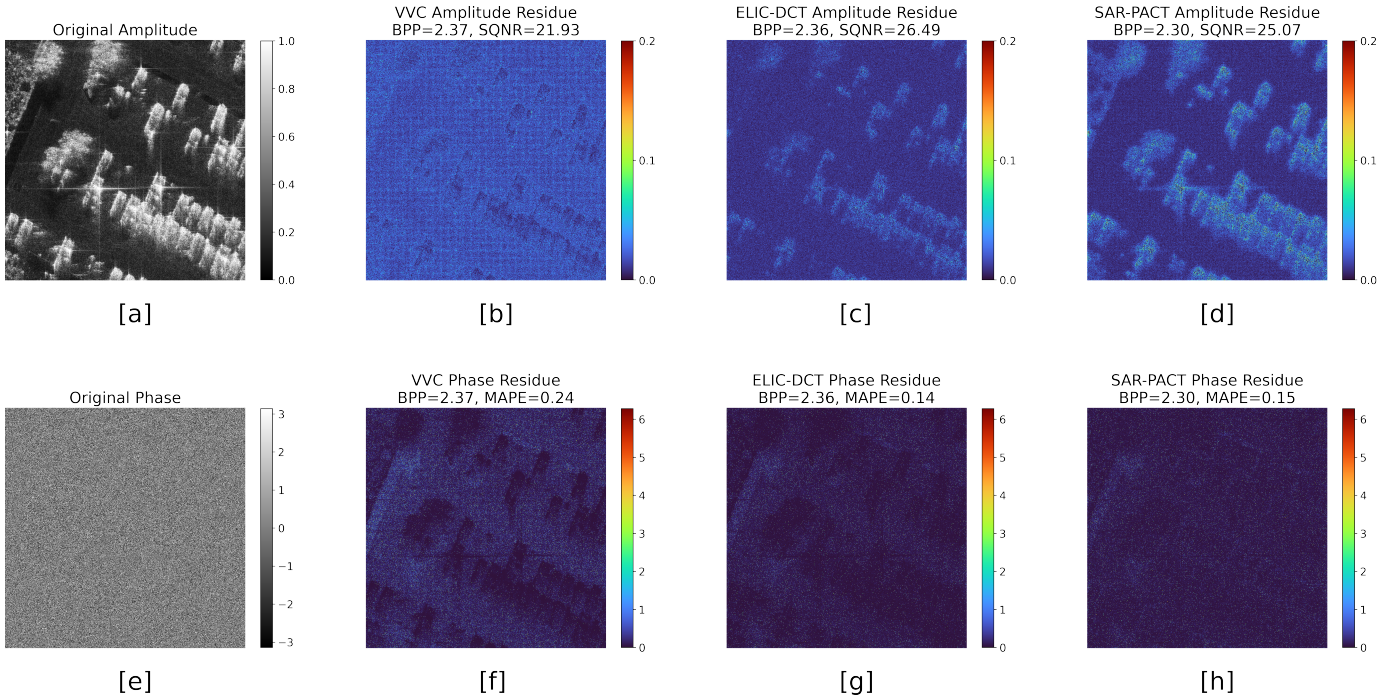


Fig. 5. Visualization of reconstructed Amplitude and Phase residues for 1024×1024 patch from NGA test image, along with their BPP per channel, SQNR, and MAPE. [a] Original Amplitude image; [b]-[d] Absolute Amplitude errors for VVC, ELIC-DCT, and SAR-PACT; [e] Original Phase image; [f]-[h] Absolute Phase errors for VVC, ELIC-DCT, and SAR-PACT.

of complexity typically seen in LIC models, as discussed in Section IV-B.

SAR-PACT’s Amplitude SQNR performance is competitive with ELIC-DCT, with superior performance at low bitrates. On Phase reconstruction, SAR-PACT outperforms both benchmarks across the tested bitrate range. The detailed RD results are shown in Table I.

Fig. 5 shows the absolute Amplitude and Phase errors for all three methods, relative to a 1024×1024 patch from the NGA test image, where darker pixels represent smaller error and brighter pixels represent larger error. While SAR-PACT shows slightly greater error than ELIC-DCT at this high BPP setting, it is important to consider the context of model complexity and efficiency, as discussed in the next section.

B. Complexity Results

We measure the complexity of LIC models using the number of learnable parameters and the number of operations per pixel (kMACs/px) via DeepSpeed [57]. Because VVC was benchmarked using the VTM reference software, rather than

a real-world implementation optimized for our use case, we do not report estimates of its complexity or encoding/decoding times.

Table I displays the results of our complexity analysis. The SAR-PACT architecture requires 88% fewer parameters and 96% fewer operations than ELIC-DCT, demonstrating a substantial reduction in model complexity while achieving competitive Amplitude and Phase RD performance across the evaluated bitrate range. These figures translate into lower memory utilization and encoding/decoding latency, lowering the barrier for deployment on embedded or edge hardware. This suggests that energy-guided compression architectures can approach SOTA bitrate and reconstruction quality at a complexity level far more practical for real-world deployment than prior learned methods.

C. Ablation Study

Grouped convolution layers provide a simple way to reduce model complexity, often with only a marginal loss of learning capacity. A grouped convolution with $G = 1$ is identical to a

TABLE I
RESULTS FOR ELIC-DCT AND SAR-PACT ON NGA DATASET.
BD-RATES ARE CALCULATED AGAINST A VVC BASELINE. BOLD TEXT INDICATES THE BETTER RESULTS.

Model	BD-Rate ↓ (SQNR)	BD-Rate ↓ (MAPE)	Encoder kMACs/pixel	Decoder kMACs/pixel	Total kMACs/pixel	Parameters
ELIC-DCT (Maharjan 2024)	-20.60%	-21.86%	139.26	261.82	469.64	33,005,378
SAR-PACT (Ours)	-20.46%	-28.45%	9.97	10.37	20.34	3,836,396

standard convolution layer, and $G = C_{in}$ indicates a depthwise convolution. This ablation study evaluates the effect of the number of groups G on complexity and RD performance. This analysis was performed on the PACT model after its first training stage, with a distortion loss term consisting of only D_{IQ} . The three tested operating points ranged from 1.4-2.2 BPP per channel. As shown in Table II, each successive group increase reduces the layer's complexity by a proportional amount. We selected $G = 8$ for our final SAR-PACT architecture, due to its combination of RD performance and complexity reduction.

V. CONCLUSION

The complexity profile of SAR-PACT represents a dramatic departure from existing learned SAR compression methods. A detailed comparison to ELIC-DCT reveals that SAR-PACT requires 88% fewer trainable parameters and 96% fewer operations per pixel, while achieving nearly identical Amplitude reconstruction quality and superior Phase recovery across the evaluated bitrate range. These figures represent a fundamental shift in the complexity-quality operating point achievable with learned SAR compression, and they translate directly to wall-clock speedup, reduced power consumption, and a smaller memory footprint.

PACT's complexity reduction is structurally motivated: the DCT tokenization produces a model input with an exploitable energy profile, the energy-guided grouped convolution concentrates filter capacity where input variance is highest, and the asymmetric hyperprior routes latent channels through branches of varying depth, avoiding redundant computation on low-energy channels. The result is a model whose complexity is more deliberately allocated, spending operations only where they matter most.

The current implementation utilizes only the HH polarization channel; incorporating cross-polarization side information is expected to improve reconstruction fidelity without proportional increases in complexity by leveraging complementary signal content across polarimetric channels. Additionally, further development of the energy-guided transform architecture presents additional opportunities for efficiency gains, which will be explored in future work.

Taken together, these results suggest that the efficiency-quality tradeoff commonly assumed in learned image compression can be addressed via architectural choices and training procedures. SAR-PACT demonstrates that SAR-optimized LIC methods can approach SOTA compression performance at a suitable level of complexity for real-world deployment.

REFERENCES

- [1] J. Curlander and R. McDonough, *Synthetic Aperture Radar: Systems and Signal Processing*. Wiley, 1991.
- [2] F. M. Henderson and A. J. Lewis, "Principles and applications of imaging radar," *Manual of Remote Sensing*, vol. 2, 1998.
- [3] A. W. Doerry, "Representing sar complex image pixels," in *Radar Sensor Technology XX*, vol. 9829, vol. 9829. SPIE, 2016, pp. 493–503.
- [4] M. Martone, N. Gollin, P. Rizzoli, and G. Krieger, "Performance-optimized quantization for sar and insar applications," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, 2022.
- [5] G. K. Wallace, "The jpeg still picture compression standard," *IEEE Transactions on Consumer Electronics*, vol. 38, pp. xviii–xxxiv, 2 1992.

TABLE II
ABLATION STUDY ON THE IMPACT OF GROUP NUMBER IN THE ENERGY-GUIDED GROUPED CONVOLUTION LAYER. BOLD TEXT INDICATES OUR SELECTED CONFIGURATION.

Num. Groups	kMACs /pixel	BD-Rate ↓ (PSNR)	BD-Rate ↓ (SQNR)	BD-Rate ↓ (MAPE)
1	1.44	—	—	—
2	0.72	-0.31%	0.27%	0.16%
4	0.36	0.64%	0.12%	0.38%
8	0.18	-0.16%	0.22%	0.45%
16	0.09	0.36%	0.61%	1.18%
32	0.04	1.14%	0.99%	0.82%

- [6] A. Skodras, C. Christopoulos, and T. Ebrahimi, "The jpeg 2000 still image compression standard," *IEEE Signal Processing Magazine*, vol. 18, pp. 36–58, 9 2001.
- [7] A. Said and W. A. Pearlman, "A new, fast, and efficient image codec based on set partitioning in hierarchical trees," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 6, 1996.
- [8] F. Bellard, "Bpg image format," 2017. [Online]. Available: <https://bellard.org/bpg/>
- [9] J. Lainema, F. Bossen, W. J. Han, J. Min, and K. Ugur, "Intra coding of the hevc standard," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 22, 2012.
- [10] G. J. Sullivan, J. R. Ohm, W. J. Han, and T. Wiegand, "Overview of the high efficiency video coding (hevc) standard," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 22, pp. 1649–1668, 12 2012.
- [11] J. Pfaff, A. Filippov, S. Liu, X. Zhao, J. Chen *et al.*, "Intra prediction and mode coding in vvc," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 31, 2021.
- [12] B. Bross, Y. K. Wang, Y. Ye, S. Liu, J. Chen *et al.*, "Overview of the versatile video coding (vvc) standard and its applications," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 31, pp. 3736–3764, 10 2021.
- [13] A. Wiecekowsky, J. Brandenburg, T. Hinz, C. Bartnik, V. George *et al.*, "Vvenc: An open and optimized vvc encoder implementation," in *2021 IEEE International Conference on Multimedia and Expo Workshops, ICMEW 2021*, 2021.
- [14] A. Wiecekowsky, G. Hege, C. Bartnik, C. Lehmann, C. Stoffers *et al.*, "Towards a live software decoder implementation for the upcoming versatile video coding (vvc) codec," in *Proceedings - International Conference on Image Processing, ICIP*, vol. 2020-October, 2020.
- [15] U. Benz, K. Strodl, and A. Moreira, "A comparison of several algorithms for sar raw data compression," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 33, pp. 1266–1276, 9 1995.
- [16] P. Eichel and R. W. Ives, "Compression of complex-valued sar imagery," *IEEE Transactions on Image Processing*, vol. 8, pp. 1483–1487, 10 1999.
- [17] P. Zhou and B. Zhao, "Novel scheme for sar image compression based on jpeg2000," in *ICEMI 2009 - Proceedings of 9th International Conference on Electronic Measurement and Instruments*, 2009.
- [18] X. Hou, J. Yang, G. Jiang, and X. Qian, "Complex sar image compression based on directional lifting wavelet transform with high clustering capability," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 51, pp. 527–538, 2013.
- [19] R. Kozhemiakin, S. Abramov, V. Lukin, B. Djurović, I. Djurović, and M. Simeunović, "Strategies of sar image lossy compression by jpeg2000 and spht," in *2017 6th Mediterranean Conference on Embedded Computing, MECO 2017 - Including ECYPS 2017, Proceedings*, 2017.
- [20] U. Pestel-Schiller and J. Ostermann, "Subjective evaluation of compressed sar images using jpeg and hevc intra coding: Sometimes, compression improves usability," in *15th European Radar Conference*. IEEE, 9 2018.
- [21] A. Kenge, M. S. Panse, S. Choudhari, H. Pethkar, and R. Pinto, "Wavelet transform based compression of raw sar data," in *2021 12th International Conference on Computing Communication and Networking Technologies, ICCCNT 2021*, 2021.
- [22] R. M. Asiyabi, A. Anghel, A. Focsa, M. Datcu, M. Martone *et al.*, "On

- the use of jpeg2000 for sar raw data compression,” in *15th European Conference on Synthetic Aperture Radar*. VDE, 4 2024, pp. 249–253.
- [23] J. Ballé, V. Laparra, and E. P. Simoncelli, “End-to-end optimized image compression,” in *5th International Conference on Learning Representations*, 11 2016, read for background on entropy modeling.
 - [24] J. Ballé, D. Minnen, S. Singh, S. J. Hwang, and N. Johnston, “Variational image compression with a scale hyperprior,” in *International Conference on Learning Representations*, 5 2018.
 - [25] D. Minnen, J. Ballé, and G. Toderici, “Joint autoregressive and hierarchical priors for learned image compression,” in *32nd International Conference on Neural Information Processing Systems*, vol. 2018-December, 12 2018, pp. 10 794–10 803.
 - [26] D. Minnen and S. Singh, “Channel-wise autoregressive entropy models for learned image compression,” in *International Conference on Image Processing*, vol. 2020-October. IEEE, 9 2020.
 - [27] Q. Xu, Y. Xiang, Z. Di, Y. Fan, Q. Feng *et al.*, “Synthetic aperture radar image compression based on a variational autoencoder,” *IEEE Geoscience and Remote Sensing Letters*, vol. 19, 2022.
 - [28] Z. Di, X. Chen, Q. Wu, J. Shi, Q. Feng, and Y. Fan, “Learned compression framework with pyramidal features and quality enhancement for sar images,” *IEEE Geoscience and Remote Sensing Letters*, vol. 19, 2022.
 - [29] L. Zhang, T. Pan, Y. Huang, L. Qu, and Y. Liu, “Sar image compression using discretized gaussian adaptive model and generalized subtractive normalization,” *IEEE Geoscience and Remote Sensing Letters*, vol. 19, 2022.
 - [30] C. Fu, B. Du, and L. Zhang, “Sar image compression based on multi-resblock and global context,” *IEEE Geoscience and Remote Sensing Letters*, vol. 20, 2023.
 - [31] P. Maharjan, C. Marrs, and Z. Li, “End-to-end compression of complex-valued sar images,” in *IEEE 26th International Workshop on Multimedia Signal Processing*. IEEE, 10 2024.
 - [32] P. Maharjan, Z. Li, and C. McGuiness, “Amplitude, phase, and gradient recovery from compressed sar images,” *IEEE Transactions on Circuits and Systems for Video Technology*, pp. 1–1, 1 2026.
 - [33] F. Pakdaman and M. Gabbouj, “Comprehensive complexity assessment of emerging learned image compression on cpu and gpu,” 12 2022.
 - [34] D. Minnen and N. Johnston, “Advancing the rate-distortion-computation frontier for neural image compression,” in *IEEE International Conference on Image Processing*. IEEE, 10 2023.
 - [35] Y. Zhang, Z. Duan, and F. Zhu, “On efficient neural network architectures for image compression,” in *IEEE International Conference on Image Processing*. IEEE, 10 2024.
 - [36] Y. Gao, R. Feng, Z. Guo, and Z. Chen, “Exploring the rate-distortion-complexity optimization in neural image compression,” *Journal of Visual Communication and Image Representation*, vol. 105, 12 2024.
 - [37] G. H. Wang, J. Li, B. Li, and Y. Lu, “Evc: Towards real-time neural image compression with mask decay,” in *11th International Conference on Learning Representations, ICLR 2023*, 2023.
 - [38] Y. Bao, W. Tan, C. Jia, M. Li, Y. Liang, and Y. Tian, “Shiftllic: Lightweight learned image compression with spatial-channel shift operations,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 35, 2025.
 - [39] M. Muckley, M. Havasi, J. Verbeek, and K. Ullrich, “Architecture optimizations for improving neural image compression compute complexity,” in *Data Compression Conference Proceedings*, 2025.
 - [40] H. Zhang, Y. Li, L. Li, and D. Liu, “Learning switchable priors for neural image compression,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 35, 2025.
 - [41] Z. Jia, B. Li, J. Li, W. Xie, L. Qi *et al.*, “Towards practical real-time neural video compression,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2 2025.
 - [42] D. He, Z. Yang, W. Peng, R. Ma, H. Qin, and Y. Wang, “Elic: Efficient learned image compression with unevenly grouped space-channel contextual adaptive coding,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, vol. 2022-June, 3 2022, pp. 5708–5717.
 - [43] A. B. Koyuncu, P. Jia, A. Boev, E. Alshina, and E. Steinbach, “Efficient contextformer: Spatio-channel window attention for fast context modeling in learned image compression,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 34, pp. 7498–7511, 6 2023.
 - [44] W. Jiang, J. Yang, Y. Zhai, F. Gao, and R. Wang, “Mlic++: Linear complexity multi-reference entropy modeling for learned image compression,” *ACM Transactions on Multimedia Computing, Communications and Applications*, vol. 21, 2025.
 - [45] Y. Li, H. Zhang, L. Li, and D. Liu, “Learned image compression with hierarchical progressive context modeling,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.
 - [46] Y. Yang and S. Mandt, “Computationally-efficient neural image compression with shallow decoders,” in *IEEE/CVF International Conference on Computer Vision*. IEEE, 10 2023.
 - [47] S. Wang, Z. Cheng, D. Feng, G. Lu, L. Song, and W. Zhang, “Asymllic: Asymmetric lightweight learned image compression,” in *2024 IEEE International Conference on Visual Communications and Image Processing, VCIP 2024*, 2024.
 - [48] J. Chen, S. H. Kao, H. He, W. Zhuo, S. Wen *et al.*, “Run, don’t walk: Chasing higher flops for faster neural networks,” in *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, vol. 2023-June, 2023.
 - [49] Q. He, Y. Li, Y. Jiang, H. Zhang, L. Li, and D. Liu, “Practical image compression with energy-guided asymmetric entropy modeling,” in *Data Compression Conference*, 2026.
 - [50] N. G.-I. A. (NGA), “Sample sicds sar dataset,” 2023. [Online]. Available: <https://github.com/ngageoint/six-library/wiki/Sample-SICDs>
 - [51] A. Krizhevsky, I. Sutskever, and G. E. Hinton, “Imagenet classification with deep convolutional neural networks,” in *Neural Information Processing Systems (NeurIPS)*, 2012.
 - [52] S. Xie, R. Girshick, P. Dollár, Z. Tu, and K. He, “Aggregated residual transformations for deep neural networks,” in *Proceedings - 30th IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017*, vol. 2017-January, 2017.
 - [53] T. K. Lee, W. J. Baddar, S. T. Kim, and Y. M. Ro, “Convolution with logarithmic filter groups for efficient shallow cnn,” *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, vol. 10704 LNCS, pp. 117–129, 7 2017.
 - [54] D. P. Kingma and J. L. Ba, “Adam: A method for stochastic optimization,” in *3rd International Conference on Learning Representations, ICLR 2015 - Conference Track Proceedings*, 2015.
 - [55] JVET, “Vtm reference software.” [Online]. Available: https://vcgit.hhi.fraunhofer.de/jvet/VVCSSoftware_VTM
 - [56] G. Bjøntegaard, “Calculation of average psnr differences between rd-curves,” in *Doc. VCEG-M33*. ITU-T SG16, 2001.
 - [57] DeepSpeedAI, “Deepspeed,” 2026. [Online]. Available: <https://github.com/deepspeedai/DeepSpeed>